

Code-Layout, Readability and Reusability

Date	15 July 2026
Team ID	XXXXXX
Project Name	FloodGuard - AI/ML Flood Prediction System
Maximum Marks	5 Marks

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform 4-space indentation throughout	Yes	PEP8-style formatting used across all modules
2	Proper File Structure	Files and folders logically organized	Yes	app/ package separates models, routes, ML logic; templates/ and static/ separated
3	Meaningful Variable Names	Variables reflect their purpose	Yes	e.g. flood_probability, feature_order, model_row
4	Function / Method Names	Descriptively named	Yes	e.g. predict_flood(), sync_ml_model_registry(), select_best_model()
5	Code Comments	Inline and block comments explain logic	Yes	Docstrings at top of every module explaining purpose
6	Modular Design	Code split into reusable functions/modules	Yes	ML logic, DB models, and routes are fully separated
7	No Redundant Code	Duplicate or unused code removed	Yes	Single source of truth for feature order (metadata.json)
8	Error Handling	Exceptions and errors handled gracefully	Yes	ModelNotReadyError, input validation, try/except around DB writes

Reusable Components / Modules

S.No	Component / Module	Language / Technology	Where Reused	Reusability Level
1	app/ml.py (predict_flood)	Python	Called from /api/predict route and the history seeding script	High
2	helpers.py table/header builders	Python (docx)	Reused across all 22 project documentation files	High
3	app/models.py (User, WeatherData, etc.)	Python (SQLAlchemy)	Reused across routes.py and seed.py	High
4	static/css/style.css design tokens	CSS custom properties	Reused across every template (base, home, predict, history)	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clear separation of concerns (app/, templates/, static/, models/, data/)
Readability	5	Descriptive names, docstrings on every module

Reusability	4	Core ML/DB logic decoupled from routes; could add unit tests for even higher reuse confidence
Documentation / Comments	5	README.md covers setup, training, and deployment end-to-end
Overall Score	4.75	Production-quality structure for a project of this scope